

MUHAMMAD ROSHAAN

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OBJECTIVE

Computer Science 3rd-year student with hands-on experience in backend development and data analytics. Skilled in FastAPI, Python, SQL, MySQL/MongoDB, and Power BI. Seeking an internship or junior role to contribute to backend or data-driven projects while gaining industry experience.

EDUCATION

Sir Syed University of Engineering and Technology, Karachi

Sep 2023 – July 2027

BS in Computer Science | CGPA: 3.81/4.00 (6th Semester)

Relevant Courses: Python, Software Engineering, Data Structures, Database Management, OOP, OS

TECHNICAL SKILLS

Backend & Architecture: Python (Advanced), FastAPI, Celery, RESTful API Design, SQLAlchemy ORM, WebSockets, JWT Authentication

Concurrency & Infrastructure: Redis (Pub/Sub, Distributed Locks, Caching), Pessimistic Locking, Asynchronous Worker Queues, Git/GitHub

Databases & Cloud: MySQL, PostgreSQL, SQLite, MongoDB, Supabase (Real-Time Sync)

Data Engineering & BI: Pandas, NumPy, Matplotlib, ETL Data Pipelines, Power BI, DAX

Core Competencies: High-Concurrency System Design, Race Condition Elimination, Zero Data-Loss Pipelines, System Stability Under Load

PRACTICAL EXPERIENCE & PROJECTS

Fleet Command Center (In Progress) | FastAPI + Redis + WebSockets | *May 2026 – Present*

- Streamed live geospatial data with sub-second latency using Redis Pub/Sub and WebSockets.
- Prevented concurrency race conditions via Redis pessimistic locking and simulated map-matched routing using OSRM.

SeatVault (Concurrency Engine) | FastAPI + Redis + Celery + React | *Mar 2026 – Apr 2026*

- Engineered a high-concurrency ticketing system using Redis distributed locks and FastAPI to prevent double-booking and race conditions during flash sales.
- Integrated Celery for asynchronous background processing and built a React frontend with real-time polling and JWT authentication.

BooknScore | Flutter + Supabase + SQLite | *Dec 2025 – Mar 2026*

- Developed an offline-first cricket scoring mobile app with real-time Supabase cloud synchronization.
- Engineered a resilient SQLite scoring engine to prevent data loss during network interruptions.

FlashSale Concurrency Engine | FastAPI + MySQL | *Oct 2025*

- Designed a high-concurrency backend system with pessimistic locking via SQLAlchemy transactions to eliminate race conditions and prevent overselling in inventory management.

CERTIFICATIONS & COURSES

Introduction to MongoDB in Python - DataCamp (*Nov 2025*): MongoDB fundamentals, NoSQL concepts, and core database operations in Python.

FastAPI (Python) - Udemy (*Jul – Oct 2025*): Built and deployed high-performance RESTful APIs with database integration, validation, and modular design (21.5 hours).

Prompt Engineering Competition - SSUET (*Jun 2026*): Participated in a university-level prompt engineering competition, applying advanced prompting techniques to AI problem statements.

LANGUAGES

English: Fluent - proficient in technical writing and professional communication.

Urdu: Native - primary language for everyday communication.